

Using Network Data to Measure Social Returns and Improve Targeting of Crime-Reduction Interventions

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All opinions are those of the authors and do not represent the position of any government organization or data provider.

Note: Estimates for school and household networks pending disclosure review (coming December 7)

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The high cost of crime and violence

- ▶ Acts of violence kill over 160 per day in the U.S. (CDC 2019)
 - ▶ And 2.3 million people live behind bars (Prison Policy Initiative 2020)
- ▶ Stark inequalities in who bears these costs
 - ▶ Violent-crime arrests 5 Xs higher for Black than White young people (OJJDP 2014)
 - ▶ With lifelong consequences on social & economic outcomes (Aizer & Doyle 2015, Mueller-Smith 2015, Nelson & Sheridan 2011)
- ▶ Social costs of gun violence alone at least \$100 billion/year, maybe > \$500 billion (Cook & Ludwig 2000, Gobbo 2023)

Randomized controlled trials (RCTs) and crime prevention

- ▶ Hundreds of RCTs over 4 decades have tested ways to address these problems
 - ▶ Policing strategies (e.g., Braga et al. 1999, 2006, 2017; Owens et al. 2018; Sherman & Berk 1984; Sherman & Weisburd 1995)
 - ▶ Employment & re-entry (e.g., Cook et al. 2015; MDRC 1980; Kemper et al. 1981; Valentine & Redcross 2015),
 - ▶ Urban environment (e.g., Branas et al. 2018; Chalfin et al. 2021)
 - ▶ Education and skill development (e.g., Armstrong et al. 2003; Dodge et al 2007; Schweinhart et al. 2007)
- ▶ Influential b/c convincingly isolate causal effect of treatment vs control condition
- ▶ Most assume one individual's behavior doesn't affect others (SUTVA)
 - ▶ Also true of many non-RCT quasi-experiments

In fact, crime is social

- ▶ 50-85% of offenders offend with others, usually as youth when most crime occurs (Conway & McCord 2002; Sarnecki 2001)
- ▶ Peer effects in crime are well established
 - ▶ Exogeneous changes in exposure to A changes B's crime (Bayer et al. 2009; Bhuller et al. 2018; Billings et al. 2019; Damm & Gorinas 2020; Drago & Galbiati 2012; Dominguez 2021; Dustmann & Landersø 2021; Norris et al. 2021; Stevenson 2017; Philippe 2017)
- ▶ But existence of peer effects $\neq$ estimate of how an intervention's crime change spreads
 - ▶ Requires knowing how many people A affects, if effect varies with who A and B are/how they're connected, if changing A's behavior or just exposure to A matters

Our paper: how do crime changes spread through networks?

► **Challenge 1: Measure social networks**

- Use administrative data on ~2m people → Chicago Police Department (arrests & reported victimizations 2005-21) and Chicago Public Schools records (2009-20)
- Co-arrest, co-victimization, same classes, same residence, geographic proximity
 - Clearly misses some strong friendship ties, but non-friend ties may matter (Granovetter 1983, Patacchini & Zenou 2008)
 - If these ties do matter, makes it feasible to look at crime spillovers retrospectively, at large scale

Our paper: how do crime changes spread through networks?

- ▶ **Challenge 2: Overcome classic identification issues** → reflection problem, endogeneity of ties, common shocks (Manski 1993, Angrist 2014)
- ▶ Leverage 4 existing, large-scale RCTs of violence-reducing interventions in Chicago (Davis & Heller 2020, Heller 2014, Heller et al. 2017, Bhatt et al 2024)
 - ▶ Given baseline tie to RCT member, exposure to T is random with known probability
 - ▶ Aronow & Samii 2017 → probability as IPW, conservative inference
 - ▶ Pooling RCTs may help solve challenges in handful of previous efforts to do this
 - ▶ Under-powered, potentially confounded (Abdul-Razzak et al, in progress, Dominguez 2023, Wood & Papachristos 2019)

What we do

1. Today: Describe networks, estimate preliminary indirect exposure effects
 - ▶ Exposure = connected to at least one peer assigned to treatment (ITT)
 - ▶ *Within RCT exposure effect*: what initial ITTs missed for original samples
 - ▶ *Out of RCT exposure effect*: impact on those not in original RCTs
 - ▶ For now, 1st-degree peer + any exposure, no geography or overlapping networks yet
2. Future: Estimate counterfactual targeting mechanisms → optimal targeting
 - ▶ Use heterogeneity to tease out behavioral mechanisms
 - ▶ Build model incorporating effects on future tie formation, network position

Intervention spillovers and social networks

- ▶ Multiple literatures developed methods to analyze peer effects, network diffusion
 - ▶ In development: how innovations diffuse through social networks, change network **structure** (Banerjee et al. 2013, Beaman et al. 2020, Beaman & Magruer 2012, Bhattacharya et al. 2013, Breza & Chandrasekhar 2019, Cai et al. 2015, Comola & Prina 2014, Feigenberg et al. 2013, Miguel & Kremer 2007, Miller & Mobarak 2014, Oster & Thornton 2012)
 - ▶ In education: how exposure to different peers or treating peers matters for **learning/behavior** (Babcock & Hartman 2010, Chaisemartin & Navarrete 2020, Dinarte & Egana-del Sol 2022, Paluck et al. 2016, Sacerdote 2001, review in Sacerdote 2011)
- ▶ Empirical crime literature: more focused on *you* when someone you know/live with gets (un)incarcerated, your school peers or cellmates change, you move, etc.
 - ▶ Exception: bullying interventions via reported friend networks (Hu 2024; Paluck, Shepherd & Aronow 2016)

What we are learning

- ▶ These networks capture many relationships
 - ▶ ~550 people exposed for every 100 people treated
- ▶ Peers of study members are different
 - ▶ Both from the study population and across network types
- ▶ Spillovers concentrated among peers also in the original RCTs
 - ▶ Observable and unobservable selection into the RCTs
- ▶ Spillovers vary by type of network connection
 - ▶ Diffusion among denser networks, substitution among closer relationships
- ▶ Spillovers are large
 - ▶ Current estimates suggest original RCTs understated violence ↓ by 40-80%

Random variation: 4 violence-reducing RCTs in Chicago

1. **Becoming a Man** (BAM 2009-10, $n = 2,740$)

- ▶ School-based cognitive behavioral therapy (CBT) intervention for 7th-10th grade boys
- ▶ ITT/LATE: Violent-crime arrests $\downarrow$ 21/45% in year 1

2 & 3. **One Summer Chicago Plus** (OSC+ 2012 $n = 1,634$, OSC+ 2013 = 5,216)

- ▶ SYEP+CBT w/ school-based recruiting in 2012, broader reach into legal system in 2013
- ▶ Violent-crime arrests $\downarrow$ 21-41% / 32-42% in year 1
- ▶ Not just incapacitation, continued decline post-program

4. **Rapid Employment & Development Initiative** (READI 2016-21, $n = 2,456$)

- ▶ 18m job, CBT, wraparound services for men (18+) at highest risk of shooting/being shot
- ▶ After 20m, 45/65% $\downarrow$ shooting & homicide arrests (adj. $p=0.13$), 38/48% $\downarrow$ S&H victimization for pre-specified subgroup

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Outcomes

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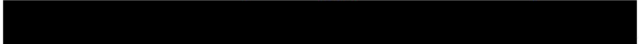
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Why a shock to criminal behavior might spill over

1. **Diffusion of behavioral change:** information, changes in attitudes/beliefs/time use get transmitted → peer crime ↓
(e.g. Akerlof & Kranton 2010, Patacchini & Zenou 2009)
2. **Key players:** lose key member of criminal team, social influencer → peer crime ↓
(e.g. Lindquist & Zenou 2014, Tankard & Paluck 2016)
3. **Crime as function of opportunity:** if one youth desists, another may take his place → peer crime ↑ (e.g. Cook 1986)
4. **Skill complementarity:** co-offending improves productivity → peer crime ↓, or perhaps substitution to new peer with similar skills ↑
(e.g. Tremblay 1993; Weerman 2003)

Constructing networks

1. **Co-arrest:** Arrested together or for same incident in 5 years prior to randomization
 - ▶ 42% in network, 2.8 ties/RCT member ($6.8 | co - arrest > 0$)
2. **Co-victimization:** Victimized in same incident in 5 years prior to randomization
 - ▶ 15% in network, 0.27 ties/RCT member ($1.8 | co - victim > 0$)
3. **Household:** Share parent/guardian name, address & last name in CPS data
 - ▶    [Details](#)
 - ▶ Limit to age > 12 for crime outcomes
4. **Shared classes:** Clustering algorithm $\rightarrow$ many shared courses in prior year
 - ▶ Schedule data currently only OSC 1 & 2, BAM coming, READI schooling often old
 - ▶    [Details](#)
 - ▶ Students in community share avg of  courses,  outside community
5. **Neighborhood** (not yet)
 - ▶ Using CPS and CPD addresses, live within m mile radius in prior year

Size and overlap of the networks: first-degree peers



RCT study sample descriptive statistics

	BAM/OSC	READI
	In-RCT	In-RCT
N	9,590	2,456
Age	17.4	25.7
Black	0.85	0.96
Male	0.90	1.00
Prior Arrests	2.5	16.7
Share Co-Arrests	0.21	0.32
Degree	2.23	5.02

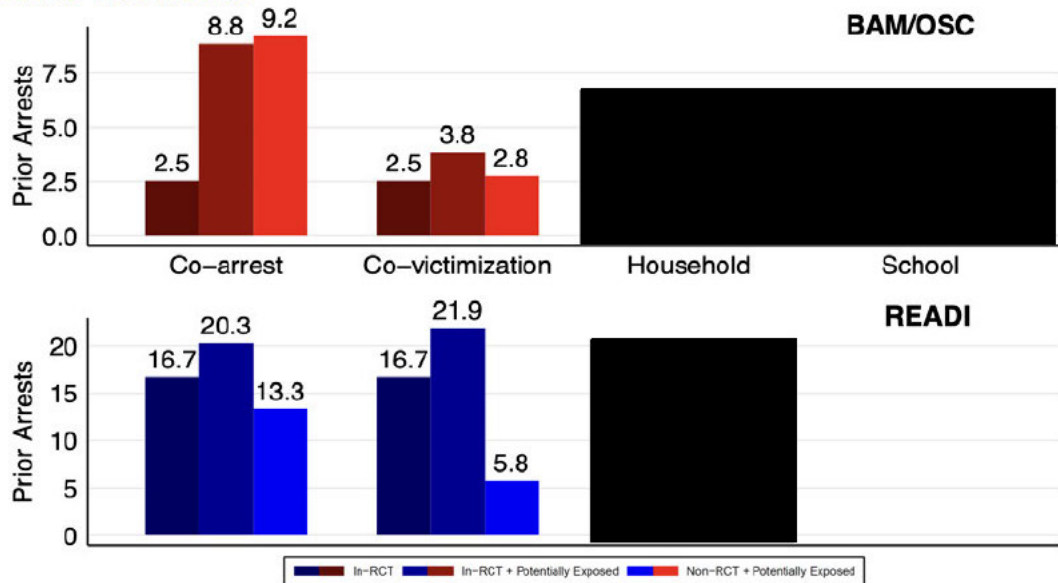
Subset in co-arrest network & tied to another study member

	BAM/OSC		READI	
	In-RCT	In-RCT + Potentially Exposed	In-RCT	In-RCT + Potentially Exposed
N	9,590	1,396	2,456	975
Age	17.4	18.0	25.7	23.6
Black	0.85	0.91	0.96	1.00
Male	0.90	0.98	1.00	1.00
Prior Arrests	2.5	8.8	16.7	20.3
Share Co-Arrests	0.21	0.63	0.32	0.45
Degree	2.23	9.56	5.02	9.18

Co-arrest linkages: First degree peers are different

	BAM/OSC			READI		
	In-RCT	In-RCT + Potentially Exposed	Non-RCT + Potentially Exposed	In-RCT	In-RCT + Potentially Exposed	Non-RCT + Potentially Exposed
N	9,590	1,396	14,257	2,456	975	6,595
Age	17.4	18.0	21.0	25.7	23.6	25.3
Black	0.85	0.91	0.86	0.96	1.00	0.95
Male	0.90	0.98	0.92	1.00	1.00	0.90
Prior Arrests	2.5	8.8	9.2	16.7	20.3	13.3
Share Co-Arrests	0.21	0.63	0.64	0.32	0.45	0.55
Degree	2.23	9.56	9.02	5.02	9.18	6.38

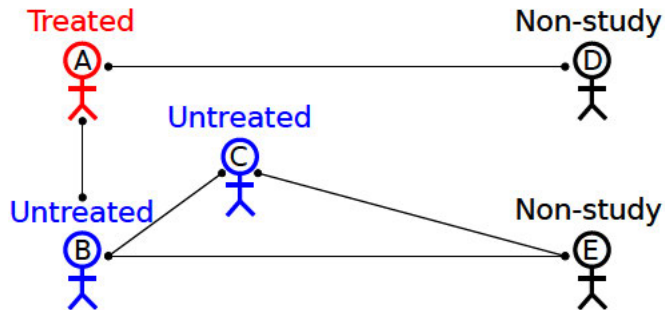
Other networks



Defining potential outcomes & treatment effects

- ▶ **If in original RCTs:** 4 potential outcomes for 4 exposure states ($\mathcal{E}$)
 $Y_{1,1}$: Directly treated, peer exposure $Y_{1,0}$: Directly treated, no peer exposure
 $Y_{0,1}$: Control, peer exposure $Y_{0,0}$: Control, no peer exposure
- ▶ Assume equal exposure effect for T and C (based on data and to help power)
 - ▶ $E(Y_{1,1}) - E(Y_{1,0}) = E(Y_{0,1}) - E(Y_{0,0})$
$$Exposure_{inRCT} = E(Y_{.,1}) - E(Y_{.,0})$$
- ▶ **If not in original RCTs:** 2 potential outcomes for 2 exposure states
 Y_1 : Peer exposure Y_0 : No peer exposure
$$Exposure_{outRCT} = E(Y_1) - E(Y_0)$$

Estimation: Aronow & Samii 2017

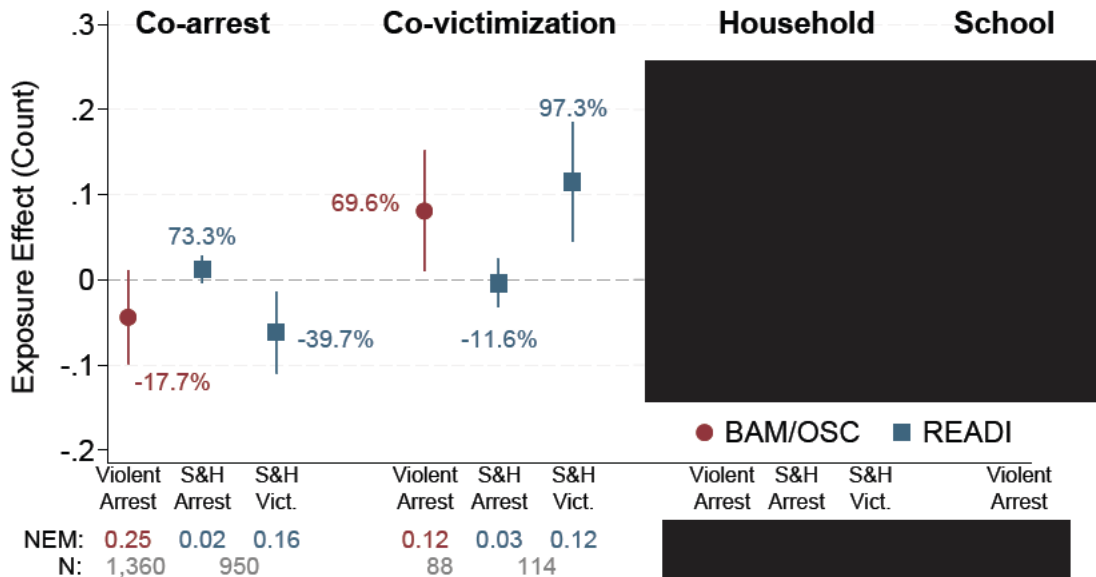


- Can calculate exact $p_{is}(j, k)$ or $p_{is}(k) = Pr(\mathcal{E} = e)$ for every exposure state

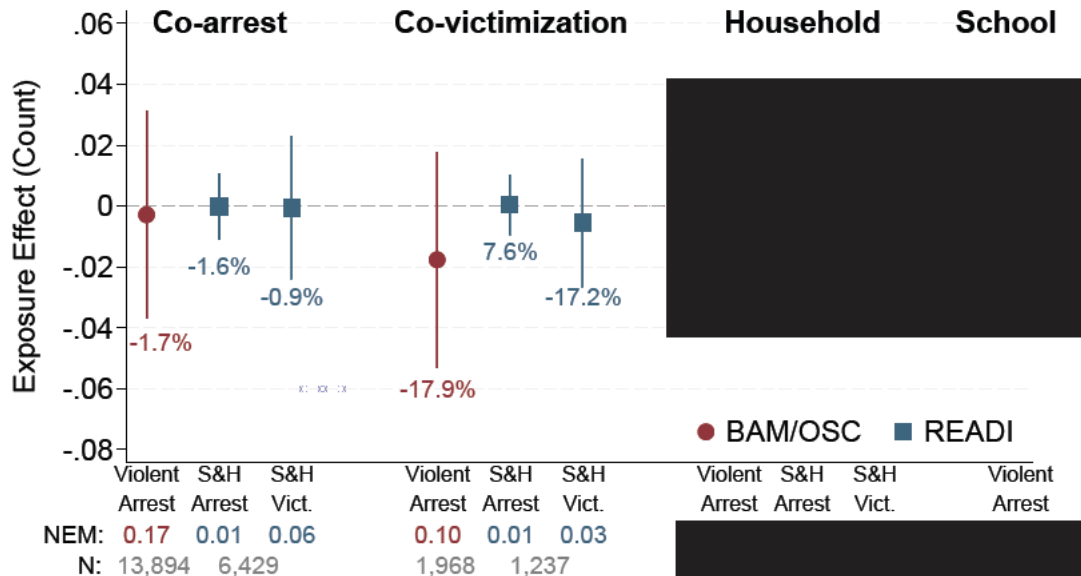
Estimation: Aronow & Samii 2017

- ▶ IPW: $E(\tilde{Y}_e) = \frac{1}{N} \sum_i \frac{\mathbb{1}(\mathcal{E}=e)Y_i}{p_{is}(\mathcal{E}=e)} \forall i$ where $p_{is}(\mathcal{E}=e) > 0$
- ▶ Estimate exposure effect = $E(\tilde{Y}_{.,1}) - E(\tilde{Y}_{.,0})$ or $E(\tilde{Y}_1) - E(\tilde{Y}_0)$
 - ▶ We do not constrain exposure effect to be the same in- and out-of-RCTs
 - ▶ Out-of-RCT peers look very different $\rightarrow$ allows for heterogeneity
 - ▶ And solves estimation problem $\rightarrow Pr(\mathcal{E}=j, k) = 0$ for out-of-RCT sample
- ▶ Residualize covariates (Aronow & Samii 2017, Särndal et al. 1992)
 - ▶ Assume $E(Y_{1,1}) - E(Y_{1,0}) = E(Y_{0,1}) - E(Y_{0,0}) \rightarrow$ residualize direct treatment
- ▶ A & S derive asymptotic standard errors accounting for interdependence & covariates, estimates are conservative
 - ▶ Assumes exposure states are correctly specified

Violence spillovers within RCT sample



Violence spillovers non-RCT sample



What could explain the heterogeneity?

- ▶ In-RCT sample observably different in at least 3 ways:
 - ▶ Higher risk of violence → non-exposed means 2-3 times those for non-RCT peers
 - ▶ More exposures → 26 vs 11% have multiple exposures, avg 1.6 vs 1.3 | exposure
 - ▶ Peers with higher take-up → 25-50% more likely to have peer who participated
- ▶ In-RCT sample unobservably different
 - ▶ RCT sample selection may succeed in identifying those responsive to change
 - ▶ Could use differences in selection across RCTs to assess

How big are these effects?

- ▶ Using estimates with $p \leq 0.1$, assume exposure effects are additive & re-calculate social impact of RCTs
- ▶ BAM/OSC: 48 RCT members indirectly exposed (in-study 1st degree peer in any network) per 100 treated people
 - ▶ Stacked ITT estimate: -2 violent-crime arrests per 100 treated (-22%)
 - ▶ Indirect exposure, all networks = -0.8 arrests per 100 treated (-20%)
 - ▶ Accounting for spillovers, average decline in violence from RCTs 40% higher
- ▶ READI: 63 exposed per 100 treated via co-arrest, co-victim, household
 - ▶ Net RCT impact shifts from -2.2 to -2.0 per 100 T for shooting & homicide arrests
 - ▶ From -1.3 (insig) to -4.0 per 100 T for shooting & homicide victimization
 - ▶ Together ~80% larger shooting & homicide decline

Summary and next steps

- ▶ Networks dense enough to really matter
 - ▶ Current estimates: original RCTs may understate net violence decline by 40-80%
- ▶ Spillovers seem limited to those in original studies
 - ▶ Maybe programs successfully targeted those at risk of violence + responsive?
- ▶ Type of social relationship seems to matter
 - ▶ Violence ↓ diffuses via denser networks, weaker ties. Closer ties → viol. substitutes
 - ▶ Potentially important lessons about joint crime decisions here
- ▶ Much work left to do
 - ▶ Geography, other exposure definitions, ties via multiple networks, school outcomes
 - ▶ Heterogeneity to inform why peer behavior matters
 - ▶ Use results, treatment effects on future tie formation to build model
 - ▶ Optimal targeting

Appendix Slides

Appendix

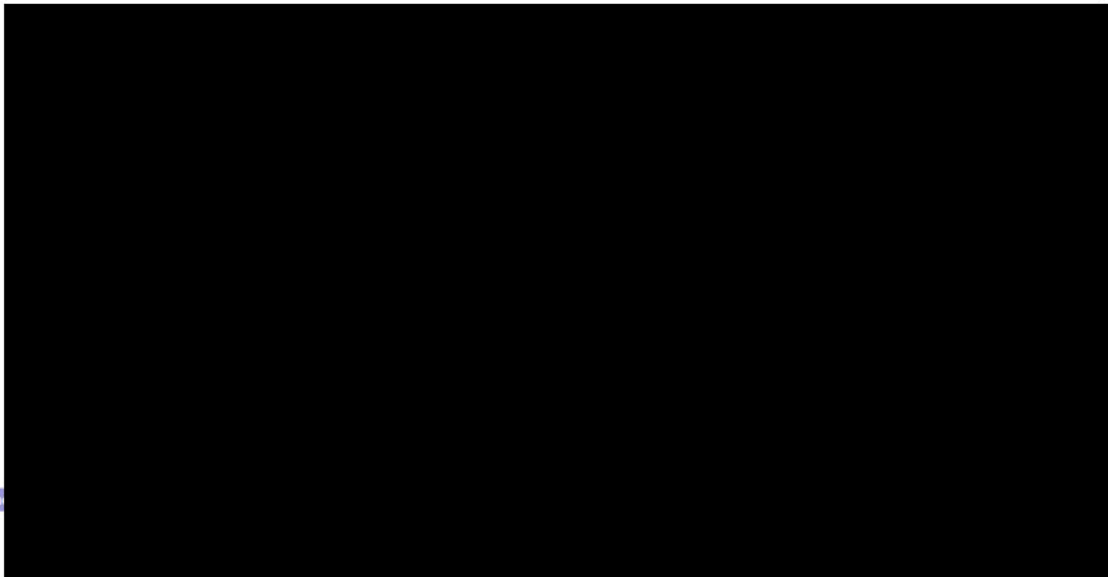
Household network

- ▶ Observe address and parent/guardian in CPS data once/year while a student
 - ▶ [REDACTED] of RCT members linked to non-missing info
- ▶ At any point in 2008/09 data (back to 90s) - randomization date, link anyone who shares:
 - ▶ Parent/guardian name, address, and last name
 - ▶ Drop co-habitants < age 12 at randomization
 - ▶ [REDACTED] linked peers are ages 6-25, some older co-residents
- ▶ Among RCT members in CPS data, [REDACTED] have sibling/co-habitant

Shared class network

- ▶ Use clustering algorithm (modularity blocking + stochastic block model) to identify “academic communities”
 - ▶ In year prior to randomization at school with most days present
- ▶ Maximizes likelihood of shared community given shared courses [Details](#)
 - ▶ Students linked if in shared academic community, not if otherwise
- ▶ Have experimented with threshold definitions (at least X shared courses)
- ▶ Schedule data currently available for OSC 1 & 2 in regular public schools (BAM coming, READI school involvement often old)

What academic communities look like



Home

Class network: stochastic block model

- ▶ Number of shared classes c between 2 students i, j function of unobserved academic network A
- ▶ Want to use data to identify peers in same academic network
- ▶ Can write down joint distribution of network membership and avg classes shared, where q is number of academic communities
 - ▶ $\log p(\mathbb{C}, \mathbb{A}) = \sum_i \sum_q A_{iq} \log \alpha_q + \sum_{i < j} \sum_{q, l} A_{iq} A_{jl} \log f_{\lambda_{ql}}(C_{ij})$
- ▶ If we knew A , could estimate α, λ via maximum likelihood
 - ▶ Instead, need to add some structure with distributional functional forms (factorized multinomial)
 - ▶ Then jointly estimate community membership and distributional parameters

Class network: stochastic block model

- ▶ Too complex to do for full data: restrict size of networks within schools
- ▶ ■ students take at least ■ of courses with students in other grades
 - ▶ So grade blocking is limiting
- ▶ Use modularity clustering algorithm – very fast (Clauset et al 2004)
 - ▶ Finding groups that maximize the difference between actual ties and what ties would look like if random
- ▶ Right now, ■ of communities have <1 shared classes with each other
 - ▶ Will likely eventually trim some “weak” communities

Non-RCT estimation in regression form

1. Control for p_{is} directly, propensity-score style:

$$Y_{is} = \tau Exposed_{is} + \delta_s p_{is} + \gamma_s + \theta X_{is} + \varepsilon_{is}$$

2. Borusyak & Hull 2021 improvement: Recenter indirect treatment by using $Exposed_{is} - p_{is}$ + randomization inference
 - ▶ BUT: when 2 kinds of treatments (direct, indirect), correct inference is on other side of frontier (not a sharp null)
 - ▶ Athey et al (2018): Fix direct T for subset & re-randomize for others – large power reductions + estimates can vary depending on whose direct T is fixed

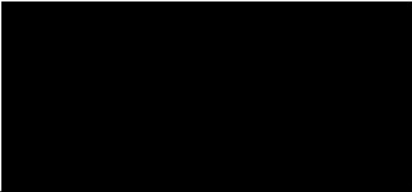
Co-victim first-degree peers

	BAM/OSC			READI		
	In-RCT	In-RCT + Potentially Exposed	Non-RCT + Potentially Exposed	In-RCT	In-RCT + Potentially Exposed	Non-RCT + Potentially Exposed
N	9,590	88	1,968	2,456	114	1,237
Age	17.4	17.5	24.4	25.7	23.9	29.9
Black	0.85	0.98	0.93	0.96	1.00	0.91
Male	0.90	0.89	0.69	1.00	1.00	0.69
Prior Arrests	2.5	3.8	2.8	16.7	21.9	5.8
Prior Victimizations	0.6	2.2	3.2	1.5	2.3	3.4
Degree	0.22	1.85	2.55	0.46	2.25	2.10

Household first-degree peers

	BAM/OSC			READI		
	In-RCT	In-RCT + Potentially Exposed	Non-RCT + Potentially Exposed	In-RCT	In-RCT + Potentially Exposed	Non-RCT + Potentially Exposed
N						
Age						
Black						
Male						
Prior Arrests						
Degree						

Classroom first-degree peers

	OSC1/OSC2		
	In-RCT	In-RCT + Potentially Exposed	Non-RCT + Potentially Exposed
N			
Age			
Black			
Male			
Prior Arrests			
Degree			